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Paleomagnetism of the Clay-Howells Carbonatite Complex: constraints on Proterozoic motion in the Kapuskasing Structural Zone, Superior Province, Canada

M.T. LEWCHUK and D.T.A. SYMONS

Department of Geology, University of Windsor, Windsor, Ont. N9B 3P4 (Canada)

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Abstract

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The Clay-Howells Complex is located approximately 130 km east of Hearst, Ontario, at 49°50′N, 82°05′W near the north end of the Kapuskasing Structural Zone (KSZ) of the Superior Province in the Canadian Shield. The pluton is a large oviform pluton of about 16 km^2 that is composed dominantly of syenite with minor carbonatite. It was emplaced into an Archean gneissic terrain of amphibolite to granulite facies with Middle Precambrian diabase dikes. The complex is unmetamorphosed and Late Precambrian in age (Rb/Sr 1075 ± 15 Ma). Multistep alternating field (AF) and thermal demagnetization was carried out on 194 specimens from 18 sites in the intrusion and three sites in the host rock dikes and gneisses. The 18 syenite sites give a unit mean direction of 294.2°, 27.1° ($N = 18$, $k = 26$, $\alpha_{95} = 7.0°$) which yields a pole position of 178.8° E, 26.5° N ($\delta_p = 4.1°$, $\delta_m = 7.6°$) for the Clay-Howells Complex. A crude contact test using the three host rock sites proved inconclusive. The pole for the intrusion is concordant, falling at about 1080 ± 10 Ma on the Keweenawan apparent polar wander path which agrees with the Rb/Sr age. Thus the complex has an untilted, primary remanence that indicates that there has been no significant uplift (< 10 km) or rotation (< 7°) on the KSZ since intrusion and therefore that all significant geotectonic activity on the KSZ must predate 1.1 Ga.

Introduction

The Superior Province of the Canadian Shield has been divided into several E-trending subprovinces. Two of these, the Quetico and Wawa subprovinces are separated from the Opatica and Abitibi subprovinces, respectively, by the NNE-trending Kapuskasing Structural Zone (KSZ) (Fig. 1). The KSZ is a 500 km linear structure that extends about 75 km to the west of the Ivanhoe Lake Cataclastic Zone (Fig. 1). It is composed mostly of high-grade metamorphic rocks with a number of early to mid-Proterozoic carbonatitic intrusions. It is characterized by pronounced gravity and magnetic highs with several normal and low-angle thrust faults. There have been a number of hypotheses for the origin of this structure such as: (a) an upwarp of the Conrad discontinuity (Wilson and Brisbin, 1965); (b) uplifted late Archean or Proterozoic horst (Bennet et al., 1967; McGlynn, 1970); (c) a suture zone between plates (Gibb and Walcott, 1971); (d) a failed arm of the Keweenawan Rift structure (Burke and Dewey, 1973); (e) a late Archean or early Proterozoic sinistral transcurrent fault (Watson, 1980); and, (f) an uplifted deep crustal slice on a low-angle reverse or listric thrust fault (Percival and Card, 1983; Hoffman, 1988). The last hypothesis is presently favoured by most researchers although the timing of the uplift and tilting on the KSZ has

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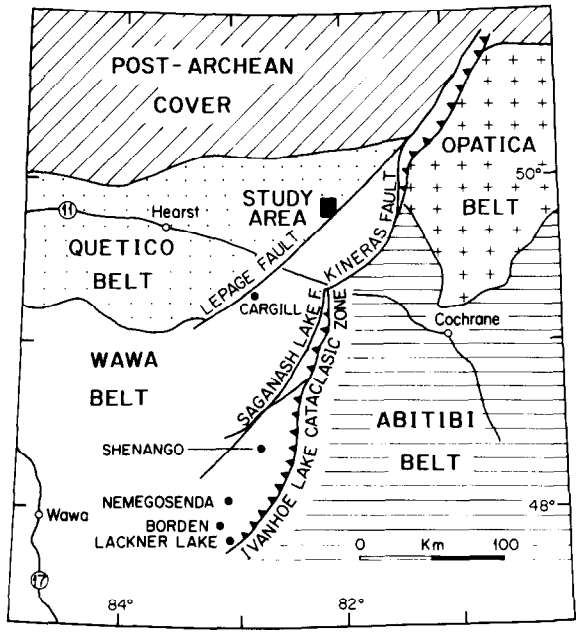


Fig. 1. Location of the Clay-Howells alkalic syenite complex in northern Ontario. The solid circles indicate other alkalic complexes in the KSZ. Geology after Leclair and Nagerl (1988).

been under investigation. Percival (1989) has postulated that the minimum age for termination of tectonic activity on the KSZ is greater than the oldest carbonatite complex that cuts faults associated with the KSZ. The oldest such intrusions are the 1888 ± 3 Ma Cargill Complex (L. Heaman, pers commun., 1988, prelim. data) and the 1872 ± 13 Ma Borden Complex (Bell et al., 1987). Hoffman (1988) proposes that the KSZ is a distant effect of the collision of the Archean Slave and Superior cratons to form the Trans-Hudson orogen (1.8–1.9 Ga) which agrees fairly well with Percival (1989). However Constanzo-Alvarez and Dunlop (1988) have suggested that reactivation at 1.1 Ga involved uplift and rifting based on paleomagnetic data from granulite gneisses and anorthosites in the KSZ.

The objective of this study was to obtain paleomagnetic data for the Clay-Howells Complex in the Quetico Subprovince and to compare its pole to the relatively well-defined apparent polar wander path (APWP) for contemporaneous Keweenawan units to look for any post-intrusive tectonic movement on the KSZ. Obviously some of the hypotheses for the origin of the KSZ involved substantial rotations and/or translations in addition to uplift that might be detected paleomagnetically.

Geology

The Clay-Howells alkalic syenite complex is located approximately 130 km east of Hearst, Ontario, at 49°50′N, 82°05′W (Fig. 1). It lies within the Quetico Subprovince of the Superior Province near the north end of the KSZ.

There is abundant outcrop along the Mattagami and Kapuskasing rivers which cut the intrusion on the north and east sides. The south side and interior are accessible by a logging road although exposure is sparse.

The intrusion is comprised dominantly of fresh unmetamorphosed pyroxene syenite. It was emplaced into an Archean gneissic terrain of upper amphibolite to granulite facies that contains N-trending Middle Precambrian diabase dikes (Sage, 1983). The complex lies along the NE-trending Lepage normal fault zone which is associated with the KSZ. A late NNW-trending fault may also cut the northeast portion of the complex. Large country rock xenolithic blocks within the intrusive body generally lack significant rotations with respect to the host rock walls, suggesting that the mode of emplacement was probably passive and the depth of emplacement was probably shallow. Field evidence and aeromagnetic survey results indicate that the attitude of the intrusion is most likely vertical (ODM-GSC, 1964). The aeromagnetic results also led Sage (1983) to suggest the presence of three distinct intrusive lobes formed by three magmatic pulses of similar composition and age. Gittins et al. (1967, recalculated using modern decay constants) obtained a Late Precambrian K–Ar age on biotite of 1004 Ma and Bell et al. (1982) obtained an older Rb–Sr whole rock age of 1075 ± 15 Ma for the intrusion.

Methods

Collecting and measurement

Five oriented core samples were obtained from each of 21 sites by field drilling (Fig. 2). These

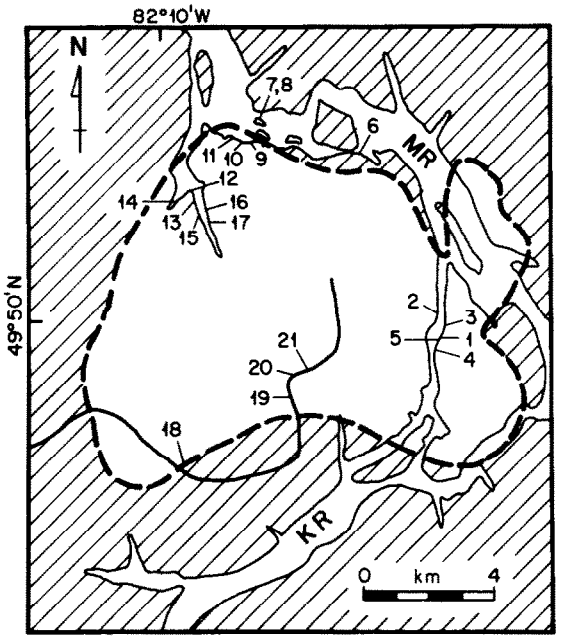


Fig. 2. Location of sampling sites in the Clay-Howells Complex. MR and KR identify the Mattagami and Kapuskasing Rivers. The dashed line is the inferred contact from geophysical data after Sage (1983). The solid line is a logging road.

include 18 sites from the complex, two from the older diabase dikes and one from the host gneisses. The cores were sliced into 2.5 cm diameter by 2.2 cm long specimens to give a total collection of 194 specimens.

The natural remanent magnetization (NRM) measurements were made using an automated two-axis Canadian Thin Film (CTF) cryogenic magnetometer operated within a magnetically shielded room to minimize viscous remanent magnetization (VRM) effects from the present Earth's magnetic field (PEMF).

Alternating field demagnetization

Two specimens from different cores for each site were selected as pilots for detailed alternating field (AF) demagnetization analysis. Each pilot specimen was demagnetized in 14 steps up to 160 mT using a Sapphire Instruments SI-4 AF demagnetizer. Normalized orthogonal vector demagnetization plots of their remanence indicate the isola-

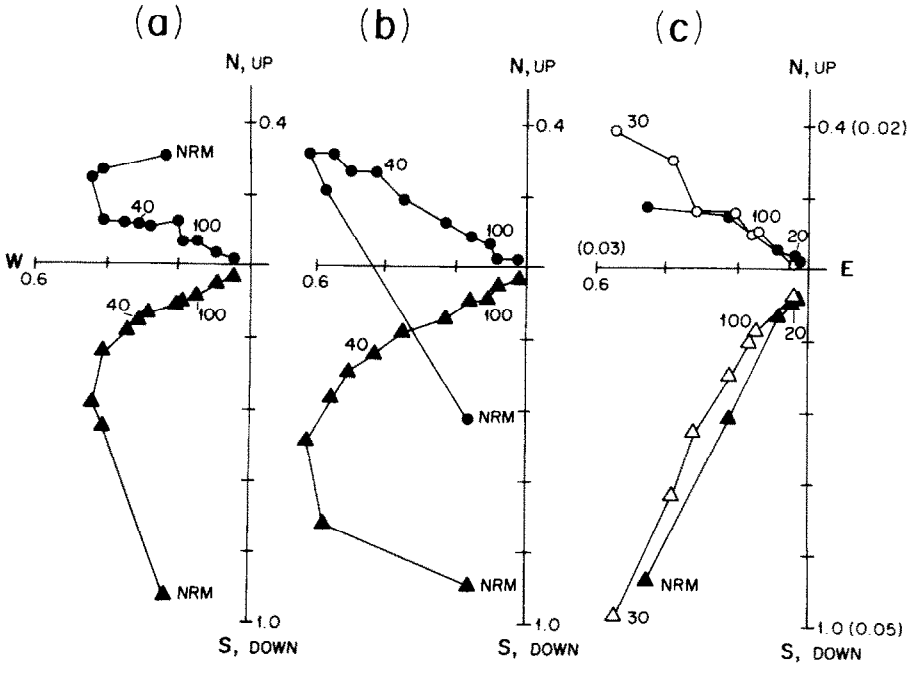


Fig. 3. Normalized orthogonal vector plots of example specimens on AF step demagnetization from: (a) site 19; (b) site 3; and (c) site 5. Treatment is in milliTesla (mT). Circles represent projections in the horizontal plane while triangles represent projections in the vertical plane. Open symbols indicate the use of an expanded axial scale as shown in brackets. Axial intensities are expressed as a ratio of the NRM intensity.

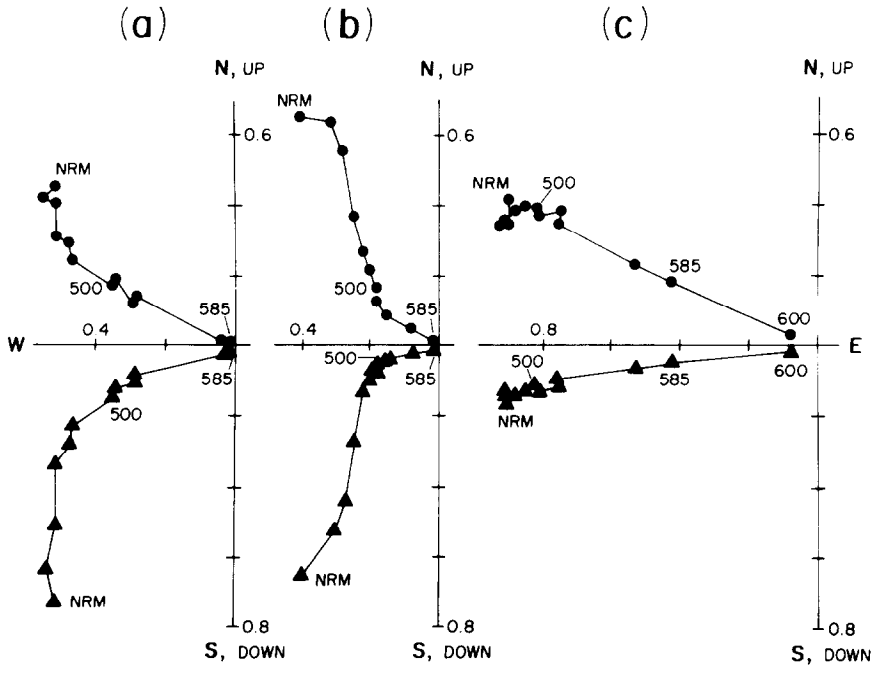


Fig. 4. Normalized orthogonal vector plots of example specimens on thermal step demagnetization from: (a) site 12; (b) site 2; and (c) site 1. Treatment is in degrees Celsius (°C). Plotting conventions as in Fig. 3.

tion of a single characteristic remanence component after removal of an initial VRM component (Fig. 3). The VRM component is generally removed by 20–30 mT of AF cleaning. Above that field the stable linear decay of a WNW moderately-dipping (20°–30°) stable component to the origin was observed in all 18 sites from the complex. The magnitude of the stable component varies from as much as 70% of the NRM intensity (Fig. 3a) to as little as 10% (Fig. 3c), indicating that the component is held by domains with a relatively wide spectrum of magnetic coercivities.

Thermal demagnetization

Two specimens from two cores that were not selected for AF step demagnetization were chosen from each site as pilots for detailed thermal demagnetization analysis. Each pilot specimen was demagnetized in 12 steps to 585°C using a Schonstedt TSD-1 thermal demagnetizer. An additional three steps to 640°C were added when deemed necessary. The normalized orthogonal vector demagnetization plots (Fig. 4) were similar to those from the AF pilots. A VRM component was usually removed by 400°C. Above about 400°C linear decay to the origin was observed as a stable component was removed. This component had the same WNW and moderately downwards direction as found for the AF pilots, and it ranged in magnitude from 70% to 10% of the NRM intensity. The few thermal pilots that required additional steps above 585°C were found to retain a small hematite component. In all cases, the hematite component direction for a given specimen did not differ significantly from its observed magnetite direction.

Bulk demagnetization

The response of the Clay-Howells specimens to both detailed AF and thermal step demagnetization methods were very similar. Thus the rest of the specimens were AF demagnetized in four steps using 5, 30, 55, and 85 mT cleaning fields for most specimens. A coherent cluster of specimen remanence directions was observed for all 18 sites which, when averaged to give the site mean directions, have cones of 95% confidence (α_{95}) (Fisher, 1953) that range from 3.5° to 13.4° (Table 1, Fig.

TABLE 1

Site and Unit Mean Directions

Site No.	N	Demagnetizing level (°C/mT)	R	Dec. (°)	Inc. (°)	α_{95}	k
1	6	580/55	5.898	301.1	12.6	9.7	49
2	11	580/55	10.624	295.5	28.8	9.0	27
3	10	580/55, 85	9.936	301.9	28.1	4.1	142
4	11	580/55, 85	10.940	280.8	15.6	3.5	168
5	8	565, 580/55, 85	7.684	309.2	43.3	12.0	22
9	10	580/55, 85	9.738	287.2	24.1	8.3	35
10	10	580/30, 55	9.904	289.7	23.6	5.0	95
11	8	565/55, 85	7.850	290.4	26.8	8.2	47
12	6	565/30, 55	5.851	293.9	24.9	11.7	34
13	6	/55, 85	5.913	298.6	45.8	8.9	57
14	6	565, 580/55, 85	5.897	292.5	25.8	9.7	49
15	10	585/55, 85	9.795	285.0	52.8	7.4	44
16	4	580/55	3.937	304.4	19.7	13.4	48
17	8	580/30, 55	7.895	289.2	22.4	6.8	67
18	9	600/30, 55, 85	8.833	296.9	22.6	7.5	48
19	10	580/55, 85	9.870	280.1	26.5	5.9	69
20	6	565/30, 55, 85	5.822	305.9	−8.9	12.9	28
21	6	580/30, 55	5.907	292.0	49.1	9.2	54
Mean	18	565–600/30–85	17.331	294.2	27.1	7.0	26

Notes: N is the number of specimens averaged for each site; R is the resultant vector obtained by the summation of the individual vectors; the mean remanence direction is given by declination (Dec.) and inclination (Inc.) in degrees; α_{95} is the radius of the cone of 95% confidence in degrees; and k is the precision parameter of Fisher (1953).

5). Averaging the 18 site mean directions gives a unit mean direction of 294.2°, 27.1° ($N = 18$, $k = 26$, $\alpha_{95} = 7.0°$). This represents a relatively reliable unit mean direction for the complex because all 18 site means were included in the analysis and because the same characteristic direction was observed over a relatively wide range of unblocking temperatures and AF coercivities. The scatter of the site means about the unit mean direction does not have a geographical bias suggesting that any post-intrusive movement along the late NNW-trending fault that cuts the complex did not produce tilting rotations. Also if three magma pulses formed the complex, they did so over a short time interval.

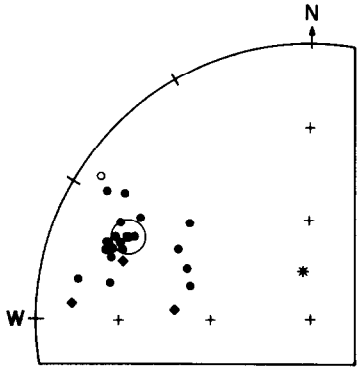


Fig. 5. Stereonet projection of site mean remanence directions. The circles represent the 18 syenite site directions, the square represents the unit mean direction circumscribed by its cone of 95% confidence after Fisher (1953) and the diamonds represent the three sites from the contact test. The star indicates the present Earth's magnetic field direction.

Contact test

A crude contact test was made using the two diabase dike sites and one host rock gneiss site, but the test proved inconclusive. The mean direction of these three sites is 278.7°, 30.2° ($N = 3$, $k = 17.6$, $\alpha_{95} = 30.3°$). Although this direction differs distinctly from the unit mean direction for the complex, its large cone of 95% confidence from having only three data points precludes a statistically significant separation of this direction from

the complex direction. The three sites used for the contact test are located much closer to the intrusion than is desirable (i.e. $< 0.1 \times$ complex diameter). The close proximity favours partial overprinting of the original remanence in these sites which would obscure their primary directions.

SIRM test

Nine specimens with representative directions and demagnetization characteristics of the collection were selected for saturation isothermal remanence magnetization (SIRM) analysis to characterize better their magnetic minerals. The specimens were initially cleaned by AF demagnetization in a field of 160 mT. They were then progressively magnetized along one axis in a continuous field (CF) in six steps to 160 mT using the Sapphire Instruments SI-4 unit. As a last magnetization step, the specimens were saturated in a 900 mT CF using a large electromagnet. The specimens were then AF demagnetized once again in five steps to 30 mT. The remanence intensities were measured along the magnetizing axis after each step.

Normalized acquisition and decay curves were plotted for the nine specimens against type curves for coarse-grained hematite, fine-grained hematite, multidomain magnetite, pseudosingle domain magnetite and single domain magnetite (Dunlop, 1971, 1972, 1973, 1981).

The acquisition curves indicate by their rapid increase in intensity that the remanence is dominantly contained in magnetite (Fig. 6a). An increase in intensity from the 160 mT step to the 900 mT step of more than about 3% is indicative of the presence of hematite. A significant amount of hematite was observed in only two of the nine specimens. Detailed examination of the initial 80 mT of the acquisition curves shows that the magnetite carriers are generally multidomain (MD) to pseudosingle domain (PSD) in character (Fig. 6b). The AF decay curves indicate most clearly that the remanence is carried mostly in MD to PSD magnetite (Fig. 6c). These results agree with the range of NRM intensity decay rates observed during the initial AF step demagnetization of the collection. The intensity decay and Curie temperatures seen on thermal step demagnetization are also consistent with the remanence being carried for the most part in MD to PSD magnetite with minor hematite in a few specimens.

Results and discussion

The Clay-Howells unit mean direction of 294.2°, 27.1° corresponds to a pole position of 178.8° E, 26.5° N ($\delta_p = 4.1°$; $\delta_m = 7.6°$). This pole

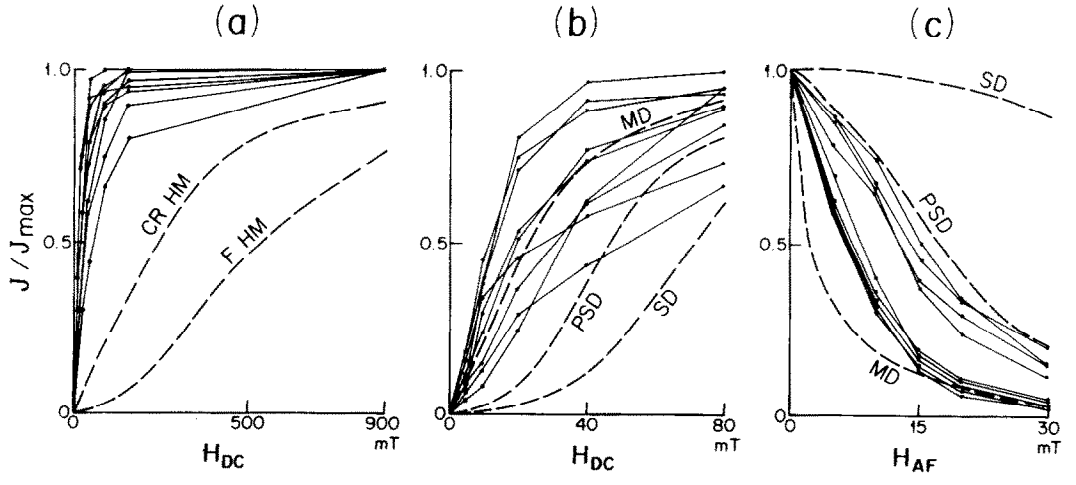


Fig. 6. Saturation isothermal remanent magnetization (SIRM) test data for: (a) acquisition of SIRM to 900 mT; (b) initial acquisition of SIRM to 80 mT; and (c) demagnetization of SIRM. The type curves for coarse-grained and fine-grained hematite (CR HM, F HM) and for multidomain (MD), pseudosingle domain (PSD) and single domain (SD) magnetite are from Dunlop (1971, 1972, 1973, 1981).

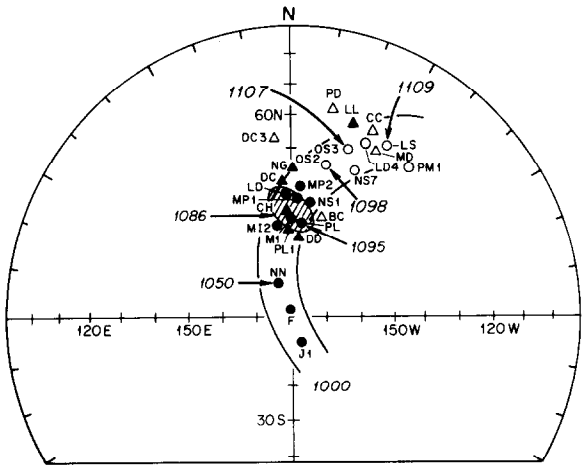


Fig. 7. Apparent polar wander path for Keweenawan time from Halls and Pesonen (1982) and Palmer and Davis (1987). The added poles are for the Clay-Howells Complex (CH — this study, with its oval of 95% confidence), Nemegosenda Complex (NG — Symons and Garber, 1974) and the Lackner Lake Complex (LL — Symons, 1988). The Powder Mill pole (PM) is that of Palmer and Halls (1986).

fits directly on the Logan Loop of the APWP for Keweenawan rocks (1000–1200 Ma) of Palmer and Davis (1987) (Fig. 7) which is probably the most intensely studied portion of the Precambrian APWP for North America. At least 60 individual paleomagnetic poles have been obtained for this time period (Halls and Pesonen, 1982). Combined with radiometric data, they provide a very good base for evaluating the age of magnetization. The Clay-Howells pole is located between the poles for the Nonesuch Shale (NN) dated at 1050 Ma (Henry et al., 1977, Halls and Pesonen, 1982) and the Upper Osler Group volcanics (OS2) (Halls, 1974) dated at 1098 Ma (Davis and Sutcliffe, 1985). It falls at the base of a dense cluster of poles that includes the Mamainse Point volcanics (MP, > 1086 Ma) and Michipicoten Island volcanics (MI, < 1086 Ma, Robertson, 1973; Palmer and Davis, 1987), the Portage Lake volcanics (PL, 1095 Ma; Paces and Davis, 1988), the Logan Dikes (LD, < 1100 Ma, Robertson and Fahrig, 1971; Pesonen, 1979) and the Upper North Shore volcanics (NS1, < 1098 Ma, Books, 1972). This places the pole for the Clay-Howells complex at about 1080 ± 10 Ma, which is in good agreement with the Rb/Sr whole rock isochron of 1075 ± 15 Ma (Bell et al., 1982).

The fact that the Clay-Howells pole is concordant to poles from contemporaneous units indicates that a tectonic correction is not warranted, and that any significant tilting or translation on the Lepage fault was probably completed before intrusion at 1.1 Ga. However, a tilt correction of up to 15° NW about the trend of the Lepage fault could be applied before the Clay-Howells pole would fall outside the accepted boundaries of the APWP, but such a correction causes the pole to move in a "younging" direction subparallel to the path towards the 1050 Ma Nonesuch Shale position. Thus this tilt correction progressively worsens the pole's fit with the Rb/Sr radiometric data and hence a correction is unwarranted. Further, SW or NE tilts of more than about 7°, caused by rotation in the fault plane, will move the Clay-Howells pole off the APWP at right angles to its trend. Therefore the Clay-Howells paleomagnetic result is consistent with the interpretations of Percival (1989) and Hoffman (1988) that all significant motion on the KSZ is pre-Keweenawan in age.

The concordance of the Clay-Howells primary component is compatible with the results from other 1.1 Ga alkalic syenite–carbonatite intrusions along the KSZ. The 1107 ± 4 Ma Nemegosenda Carbonatite complex (L. Heaman, pers. commun., 1988) records an intermediate WNW component (Symons and Garber, 1974) similar to Clay-Howells and the 1108 ± 10 Ma Lackner Lake Complex (Symons, 1988) records a somewhat older concordant Keweenawan pole position. Preliminary data on the 1047 Ma Shenango Complex (D. Dunlop, pers. commun., 1988) indicates an intermediate ESE up component that is similar but antiparallel to the Clay-Howells direction. All four of these bodies are in or along the KSZ and give concordant pole positions that do not indicate significant postintrusive translation or rotation on the KSZ. Furthermore there is no evidence from their surface outcrop, magnetic anomaly pattern, distribution of petrologic phases or internal foliation fabric that any of these complexes have been tilted since emplacement. Thus all four appear to postdate the majority of uplift and tilting associated with the formation of the KSZ. There is, however, potential evidence in the paleomagnetic data for a

small amount of post intrusive uplift along the KSZ. Geobarometric data by Percival and McGrath (1986) indicate a metamorphic pressure of 5.4 ± 0.5 kbar for the Archean granulite host rocks surrounding Clay-Howells which is equivalent to a depth of about 18 km. They also conclude that there was at least 8 km of uplift before 1.9 Ga so that the maximum emplacement depth for Clay-Howells would be about 10 km. The paleomagnetic data for Clay-Howells is consistent with this estimate. Thermal step-demagnetization of the syenites revealed a VRM component up to about 400°C and a single primary component thereafter. There was no indication of any secondary metamorphic overprinting in the collection. A 400°C laboratory heating temperature corresponds to a geological burial temperature of about 200°C (Pullaiah et al., 1975). In a stable shield environment a reasonable geothermal gradient would be about 25°C/km for the upper 10 km of the crust (Blackwell, 1971; Čermák and Jessop, 1971). Thus a 200°C burial temperature is equivalent to an emplacement depth of about 8 km and postintrusive uplift cannot have exceeded this value otherwise a secondary metamorphic overprint would have been preserved. Similarly, the other three well-studied Keweenawan intrusions in the KSZ —Nemegosenda, Lackner Lake and Shenango— also lack regional remanence overprints. This means that rifting and intrusion in the KSZ took place over a 35 Ma period from 1110 to 1075 Ma with no evidence for subsequent uplift or metamorphic activity. It is also evident that the steep reversed ENE component in the Archean host rocks that Constanzo-Alvarez and Dunlop (1988) attribute to thermochemical overprinting during Keweenawan rifting and volcanism is not found in these intrusions. Therefore it is probably related to some earlier event since the APWP does not pass through this area after 1.1 Ga.

Conclusions

The Clay-Howells alkalic syenite-carbonatite complex records a primary remanence. Although tectonic rotation ($< 7°$) cannot be ruled out entirely, our preferred interpretation is that the complex has not been tilted and that the remanence records the Earth's magnetic field at the time of intrusion at about 1080 ± 10 Ma. This result agrees with both the available radiometric dating for Clay-Howells and the paleomagnetic results from other 1.1 Ga complexes in the KSZ. Therefore translation, thrust-fault rotation, or vertical axis rotation associated with tectonic activity on the KSZ, regardless of the hypothesis involved for its origin, had ceased by prior to 1.1 Ga although about 8 km of postintrusion uplift and erosion is probable.

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